Rishi Sandrana

872-305-3172 | rishi.sandrana@gmail.com | github.com/RishiSandrana

Education

The University of Texas at Dallas Aug. 2026 – Dec. 2027
Bachelor of Science in Computer Science (expected) *GPA: 4.00*

University of Illinois Urbana-Champaign Aug. 2024 – May 2026
Computer Science coursework
Relevant Coursework: Machine Learning, Algorithms & Models of Computation, Data Structures, Probability & Statistics for CS, Database Systems

Certification: AWS Certified AI Practitioner (AIF-C01) Jul. 2026 – Jul. 2029

Experience

Associate Data Scientist Intern, AI & Advanced Analytics May 2026 – Jul. 2026
IBM *Dallas, TX*

- Automated conversion of **900+ process flow diagrams** from client documents into **BPMN 2.0** using vision agents, a deterministic processing workflow, and a **12-check validator**; integrated with **SAP Signavio via REST**, reducing manual work by **70–80%**
- Built a **LangGraph** workflow to classify backend API errors into **22 categories**, generate **Excel** failure reports, and route issues to accountable engineering teams for a **Snowflake**-backed GenAI analytics platform serving **Fortune 50** executives
- Authored **100+ Playwright regression tests** covering data quality, dataset management, access roles, and evaluation tools; replaced manual QA across development, staging, UAT, and production promotions

Climate Hub Intern Jun. 2025 – Aug. 2025
Discovery Partners Institute *Chicago, IL*

- Built a full-stack **LLM-powered** web app with **Next.js**, **Python**, and **PostgreSQL**, enabling natural-language questions and insights from synthetic healthcare and environmental datasets
- Developed **Python ETL** pipelines to clean, transform, and load Synthea patient data from **Google Drive** into **Supabase** and **Dockerized PostgreSQL** using **SQLAlchemy**

Projects

Hypershift | *Python, PyTorch, pandas, NumPy* Sep. 2026 – Present

- Developed a five-seed **THINK+mixer stock-ranking ensemble** in **PyTorch**, achieving **19.6% net CAGR versus SPY's 10.9%** in a **2022–2025 historical backtest** accounting for transaction costs and liquidity constraints
- Built a **Python** research pipeline processing **10.8M+ market-data bars** across **156 historical NASDAQ-100 securities**; validated forecast timing and ticker identity, independently reproduced model checkpoints, and reconciled **241 portfolio paths** tracking cash, holdings, and fees

Rear-End Collision Prediction | *Python, OpenCV, YOLOv4, Jupyter* Dec. 2022 – Apr. 2023

- Created a novel computer vision algorithm using **OpenCV** and **YOLOv4** to detect vehicles and predict rear-end collisions from dashcam footage in real time
- Formulated a **custom 4PL nonlinear regression model** estimating real-world vehicle speeds from pixel distance and frame time; **won Gold Award** at the Illinois Junior Academy of Science State Exhibition

CipherBot | *Python, PostgreSQL, Flask, Render* Aug. 2021 – Present

- Shipped a **Python** Discord bot automating cipher encryption and decryption through **20+ commands**, adopted by **135+ servers** with over **10,000 user interactions**
- Deployed on **Render** with uptime monitored via **UptimeRobot**, maintaining **99.4% availability** in production

Technical Skills

Data & Analytics: Python, SQL, PostgreSQL, Excel reporting, Snowflake, pandas, NumPy, SQLAlchemy, ETL, time-series forecasting, statistical modeling
AI & Automation: LangGraph, LangChain, prompt engineering, AI agents, MCP, RAG, Claude, OpenAI, PyTorch
Engineering & Tools: REST APIs, Playwright, CI/CD, Docker, Git, Supabase, Flask, Next.js, SAP Signavio
Additional Languages: Java, C++, JavaScript